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Learning as Reflective Conversation with Materials: Notes from Work in Progress

Jeanne Bamberger and Donald A. Schön

We present, here, sketches from our current work. The work, itself, has been evolving over a number of years. Its workplace has been a graduate seminar which we called "The Role of Metaphor in Learning and Design." Already in the early years of the seminar it became clear that our interest was not in metaphors (in the plural) as artifacts of language but rather in metaphor as an instance of a generative process. We were interested in what we called "generative metaphor" and the process of making a metaphor, as this expressed a moment (or more often, the momentary articulation of an on-going process) in which a person "comes to see in a new way."¹ The concern for generative metaphor has gradually evolved into a more focussed attention to the *making* process itself, and indeed, to the making of *things*, as this, too, involves coming to see in new ways. Thus, the emphasis has shifted from the "metaphor", in our course title, to "learning and design" as these latter interact in the course of making and shaping.

Our questions have also evolved. We initially posed the more traditional one: Is making something—a melody, a design for a building, a pattern of tiles—primarily what is thought to be an intellectual/"cognitive" activity like problem solving, or is it rather a special sort of activity wedded to particular mediums and materials and dependent on special skills or "talent"? But more recently we have turned the question around: Can we, from a careful and rigorous study of the process of making things, gain insights that will apply to learning and invention in domains more usually considered "cognitive"—making solutions to problems in math and science, solving puzzles within con-

strained task situations, inventing new theories? In short, how can making a real object inform making things in domains thought to be quite different from the arts and crafts?

Towards a Methodology of Research

What contributed to this turn-around? At some point, while still engaged with the initial question, we set ourselves the task of trying to capture moments in which individuals actually came to see in new ways. To do this we designed simple but rich task situations and video-taped the work of individuals doing them. Some of the tasks we borrowed from more traditional ones (like the Vygotsky block task), but some were more open-minded (like making a tune). Our real work began when we faced the problem of making sense of what our task-participants were doing.

The modes of analysis evolved over a number of tasks and also over several years, eventually raising questions that went far beyond the immediate objective of capturing moments of insight. The most powerful strategy we found as a starting point for our analysis was something we called "chunking the protocol." This involved looking for what seemed important *boundaries* that articulated observable phases or organic "chunks" within the continuing course of a participant's work. These we thought might signal shifts in behavior and/or focus in the evolution of the participant's understanding.

On the first several passes over the tape, we searched for such boundaries without trying to be explicit about the criteria we were using or exactly what sorts of behavior were signaling the boundaries we found. We simply tried to mark "something new happening." Once having found a chunking that

seemed right, we went back and looked for the criteria we had quite spontaneously used. In other words we reflected on our own behavior while at the same time letting the behavior of our participant "talk back" to us in the context of what we saw as possibly significant turning points in his/her work. We asked, given this chunking, what are we taking to be "something new" and why? What does this tell us about what we take to be the purpose of the task, or, indeed, what we take even to be "seeing" in *some* way, let alone a *new* way?

This initial chunking led to others where we now explicitly set criteria different from the ones we had found ourselves using in making the first chunking. We also showed the tape to other persons whom we expected to bring a different "set of eyes" to the protocol, working with them to help us determine the criteria *they* were spontaneously using.

These various chunkings served to help *us* see in new ways. New moves, new behavior, new features of the protocol were "liberated"—i.e., things we hadn't seen at all became "visible." But most important, we gained insight into our own, often tacit assumptions concerning the nature of the task, the structure and theory of the task-domain, and even the possible elements and relations that we had taken to be "givens" in the materials.

The result was a collection of descriptions that might resemble a series of maps, all of the same terrain, but with the delineation of points and parts and the lines and shadings that did so, all quite different depending on what the cartographer was paying attention to as significant things (topography, roads, weather). And just as each of these maps is "right" in its own terms, so we took

each of our chunkings of the protocol to be “right” in *its* own terms. The problem in our case was to be convinced that we had correctly discovered assumptions-in-use underlying our various chunkings of the protocol. For these, in turn, would at least guide what we considered a new way of seeing on the part of a participant.

In addition we needed to find ways of coordinating these multiple views so as to inform our ultimate search. But with multiple views available, we had a better chance of answering our initial question: What was the nature of the processes that led our participants to *learn*, in particular, to learn *through their own experience*, new knowledge that was not present at the outset of their work?

Plato puts this problem of learning “on your own steam” and recognizing “it” when you find it, into the mouth of Meno (in the dialogue of that name):

But how will you look for something when you don't in the least know what it is? How on earth are you going to set up the object of your search? To put it another way, even if you come right up against it, how will you know that what you have found is the thing you didn't know? (Meno 80.D)

And after much discussion, Socrates says (in a tribute to “discovery learning”):

... one thing I am ready to fight for as long as I can in word and act: that is, that we shall be better, braver and more active men if we believe it right to look for what we don't know than if we believe there is no point in looking because what we don't know we can never discover. (Meno 86.C)

This process we were able to see among our participants: Looking for something they did not yet know, they most often found it. And most often it emerged through a “piton effect,” like the process of pulling yourself up through your own power to a new position (and view) on a mountain. The question then becomes, what are the pitons that our participants used in the course of their work to achieve a new view of their materials and tasks? It was, to pursue our metaphor, because of the nature of the pitons we found that we turned from thinking of “making” as a cognitive process to a view of cognitive processes as a kind of “making.” We found our participants improvising;



“In short, how can making a real object inform making things in domains thought to be quite different from the arts and crafts?”

we found them engaged in on-the-spot experimenting in response to the new phenomena they were discovering. In short, we found them “conversing” with their materials. Their conversations were more like the making and shaping of coherence in the arts than like the means-ends, instrumental logic usually associated with puzzles and problems. The example to which we will turn in a moment illustrates the rather unexpected nature of the events that seemed to lead our participants beyond what they knew and even to recognize what they found as “the thing they didn't know.”

In order to capture the sense of movement and instability associated with, even necessary to, learning and change, we have coined the term, *knowledge-in-action* (KIA). By KIA we mean the current state of an individual's possible mental constructions for shaping coherence with respect to some present phenomena. We use the term in place of more traditional expressions such as “internal representation” or “cognitive schemas” in order to capture this sense of continuing mobility. We also want to suggest by the term that KIA need not be associated with a capacity for external symbolic expression. Thus KIA may often be that which an individual knows how to *do* but cannot yet *say*.

However, we also intend to include in KIA, individuals' current capacities for making descriptions in the given domain as well as in others. This because there are often important interactions between individuals' KIA with respect to making descriptions and their current capa-

cities to act on, recognize, and construct coherence within the materials of a present situation. This interaction between materials and modes of description will in fact play a major role in what we have called the “piton effect.”

Conversational Learning

The task we have chosen involves two musically untrained adults in the work of making a tune. The two participants (who worked together on the task) were given five Montessori bells² as their building materials. The particular bells/pitches were selected by us beforehand so as to make the task intriguing and also somewhat problematic (more on this, below). The instructions, as they were actually given to the pair, were as follows: 1) make a tune that you like using all the bells in the collection you have been given; 2) make as rich a description as possible of your completed tune; 3) write a set of instructions so someone else could play your tune on your bells. The task was, thus, open-ended in that there was no “right answer” and particularly because the criteria for success depended on the participants, themselves. We should say right off that these participants (and others after them) were all able to build tunes that, in fact, made sense to them as well as to others. Thus their criteria for coherence went beyond mere personal “taste.” Our interest, then, is not in whether or not the participants were able to complete the task, or whether they could do it at all (they all could), but rather the evolutionary course of their work—the moves they made, the strategies they used, and most of all the ways in which general, shared criteria for coherence emerged.

We have chosen this protocol because it seems a paradigmatic example of what we have come to call “conversational learning.” By this we mean the gradual evolution of making something through reflective “conversation” between makers and their materials in the course of shaping meaning and coherence. “Reflective” has at least two intentions here, and often they are so intertwined as to be indistinguishable: the makers' spontaneous (and active) reflective response to their actions on the materials, and the “reflection” of the materials (as they take various shapes and forms) back to the makers. The two kinds of reflection can be thought of as two kinds of “talking back.” In the first, the

makers talk back to the materials (re-shaping them); in the second, the materials talk back to the makers, re-shaping what the makers know about them. The distinction is, in a sense, moot since materials don't "talk" and the "talk" of the makers is most often (but not always) action rather than words. Further, the back-talk of the materials is only to the extent that the makers "hear" it—i.e., the current state of the makers' knowledge-in-action strongly conditions what they recognize and apprehend as the "message" implicit in the

“...as the participants shift in their uses of the materials and with these shifts also in their modes of description, new features and relations are ‘liberated.’”

current state of the materials. In turn, the makers' talk back to the materials is not "heard" by the materials except to the extent that these are re-shaped in some way.

The metaphor of "conversation" can serve a two-fold purpose: it can, as suggested above, serve as a way of viewing the evolving course of our participants' work; at the same time, it can serve as a way of setting a mood, an ambiance for research. In this latter sense, the notion of "conversation" becomes important and productive as it encourages us, as researchers, to make our own action experiments in search of the emergence of new meaning, new features, new structures. For example, as in the process of multiple chunkings of a protocol discussed earlier this means being vulnerable to the possibilities of restructuring our assumptions: what we take to be the "givens" of the materials and indeed, the "givens" implicit in the theory of the domain. At the same time, such an approach requires that we meet the challenge of rigorously testing the validity of emerging new structures against

the observable evidence of the protocol.

The question of evidence becomes central since what constitutes evidence is necessarily influenced by the knowledge-in-action that we, ourselves, bring to the situation. We are, then, sensitive to the problems of the "hermeneutic circle"—that is, our interpretations of the protocol as "text" are dependent on what we are able to see, the filters built in to our apprehension; the participants' interpretation of the task, its materials, and its criteria for success are also dependent on what they are able to see, the filters built into *their* apprehension. At worst this sensitivity results in despair, something like the despair expressed by Meno. We aim then at something more like a "hermeneutic spiral" which moves dialectically through possible interpretations moving out and beyond, rather than a closed circle where head and tail forever meet going nowhere.

Making Meaning

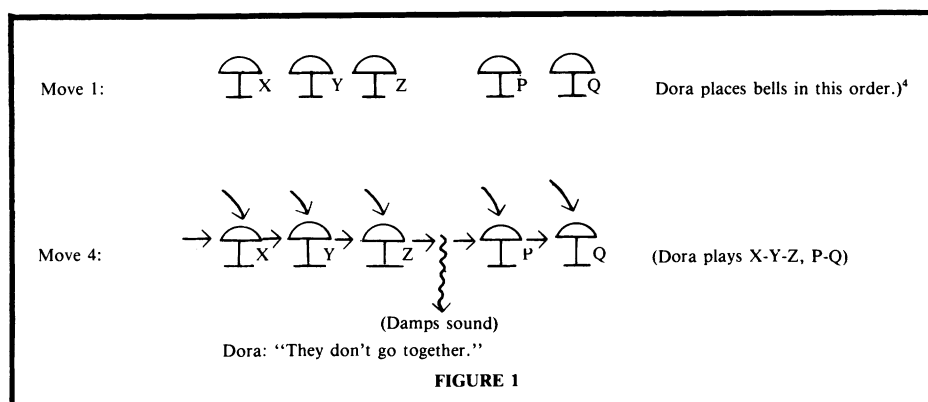
If we take seriously the notion that meaning, itself, is a process of making, describing the work of the participants presents certain difficulties at the outset. For example, if we name the pitches of the bells the participants were given with their conventional letter names (D, G, etc.), the reader is already privy to information that the tune builders were not. More importantly, such information is quite different in kind from that of the players, including a whole set of underlying assumptions that we cannot appropriately attribute to the participants as they begin their work. More importantly, giving the reader the names of the bells/pitches along with the meanings they carry, puts us in danger of reading back onto the participants' moves—their decisions and actions—"givens" that are ours but not theirs. At the same time we run the risk of failing to recognize and to appreciate what it is the participants *can* do and the cognitive work involved in their on-the-spot constructions and decisions.

The problem points up the powerful role of learned and internalized structures associated with symbols and categories of analysis as these influence the way the "informed observer" actually comes to see, and in this case, to hear. In short, it points up the kind of problem we faced in *our* continuing "conversations" with the data of the protocol.

Recall that unlike musically trained

tune-builders, our untrained participants have no way of placing and thus naming the bells/pitches with respect to a constant, fixed reference such as a scale, a key, or units of measure for identifying the intervals among and between them. As a result, the participants must *give* meaning to the elements by making a coherent universe within which they *gain* meaning—a universe that can include them. For example, while the goal of the participants is to make a tune, the evolution towards this goal includes making a number of "transitional objects"—namely a series of constructed and reconstructed bell-arrangements on the table. These transitional objects in their various transformations serve as the intermediary between the process of shaping coherence, itself, and the final tune. At the same time the arrangements of bells on the table also serve to "hold still" the meanings the participants give to the bells. Each arrangement becomes a reference for these meanings. And since each arrangement is also a concrete entity, we call each transitional object a *reference entity*—an embodied and enacted description of what the participants know so far. The meaning and functions of the bells/pitches evolve in reciprocal interaction with the construction of the reference entities. As such they serve as a materialized "log" of the making process—a series of sketches.

The term reference entity is meant to contrast with *reference structure* which we take, for example, to be available to musically trained individuals. By reference structure we mean that complex network of internalized mental structures which is closely linked to the conventional symbol systems associated with a domain—here, music. We propose that conventional symbols serve as the intermediary between such internalized mental reference structures and the stuff of the domain. Indeed, we would argue that the notations determine, in important ways, just what are taken to be *possible* entities and relations in the stuff. In turn, the conventions of notation play a significant role in the development and the nature of such internalized reference structures. Thus, there is a reciprocal interaction between the ontology implicit in the notation system, the ontology inherent in the reference structures, on the one hand, and, on the other, what is taken to exist as phenomena in the domain. In the context of a



reference structure, then, names name properties or relations represented in internal reference structures and found by instantiation, in the phenomena of concern.

By contrast, the type and function of naming is noticeably different in the case of a reference entity. A reference entity serves a naming *function*, but it does not literally name. That is, a reference entity is unique, often transient, and it is "held" by the materials used for making things within the domain. A reference entity serves to single out, externalize, and hold for current attention some emergent object or relation. For example, the position of a bell in the array can, for the builder, serve as a way of referring—most often as a way of referring to a kind of thing that is not defined with respect to its explicit inner properties or relations. The position of a bell, or indeed, a whole ordering of bells, functions, at some moment, to "dub"³—to stand for, call up, or point to—some relation or property not yet differentiated or defined but *recognized*. Further, the same positioning can refer at *this* moment to one property, at *that* moment, to another. Thus, a reference entity serves the function of on-the-spot naming within, and as part of, the making process.

Snapshots from the Protocol

We include, here, just three snapshots, or better, three "stills" from the participants' work. All of them are from relatively early in its course: moves 1 and 4, moves 8-10 occurred in the first 5 minutes; moves 27-30 occurred about five minutes later in this 45 minute protocol. Each of these glimpses share important characteristics: the reference entity is transformed in its use, in what it describes, and/or in its shape. In addition each example includes several

media of description—space, gesture, sound/time, words.

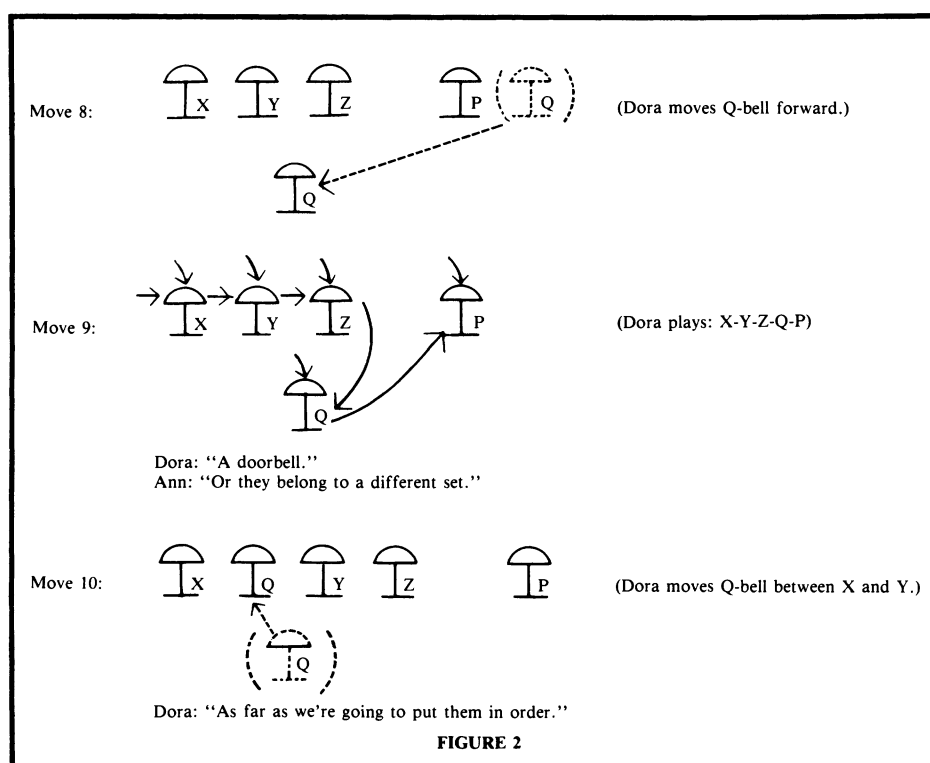
In these first moments of the participants' work we see them engaged in a series of continuing experiments. Their questions seem to be: What have we got here; what sense can we make with it? We will argue that as the participants shift in their uses of the materials and with these shifts also in their modes of description, new features and relations are "liberated."

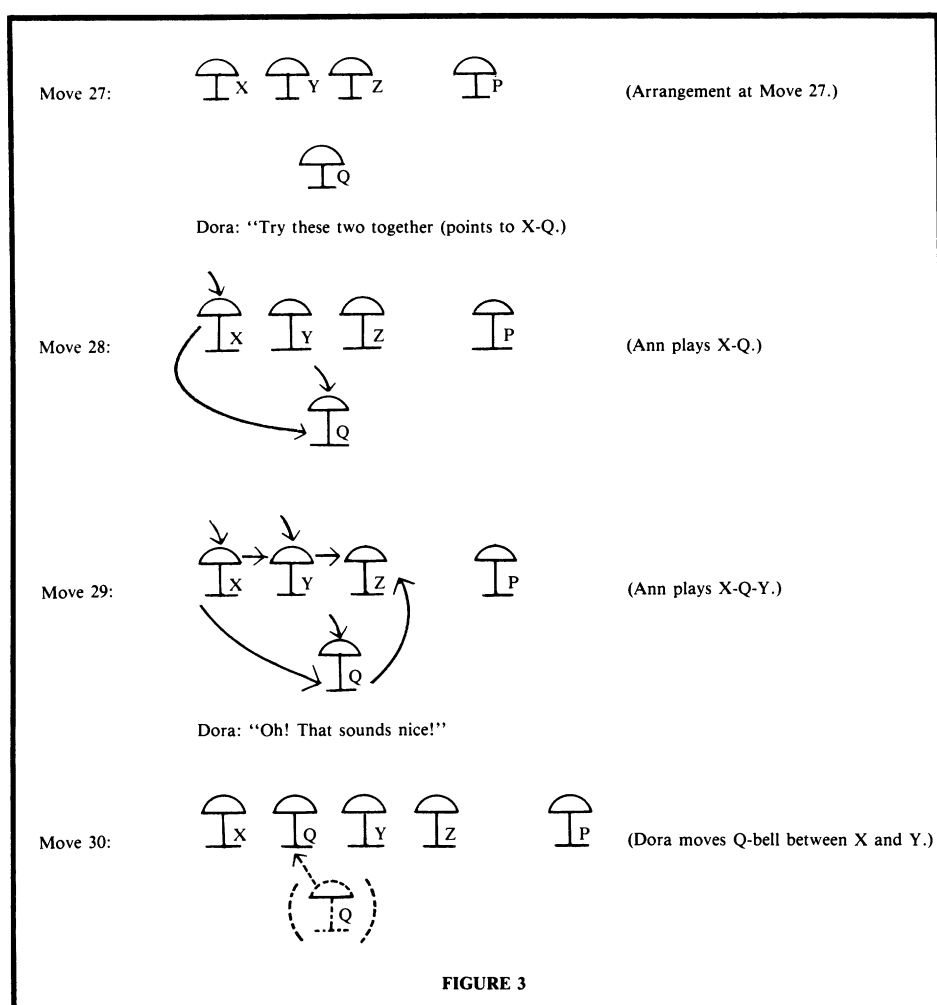
The tape begins after Dora has been at work by herself for a few moments casually experimenting with the bells. At Move 1, Dora, using the materials as objects to be moved about in space, makes a pattern consisting of two groups of bells. The inner ordering of the bells and

the space between them enactively describe the results of her initial experimenting. Dora has made a first reference entity, holding still what she has found so far: there are two *kinds* of bells—X Y Z is one kind, P Q another.

At Move 4 Dora makes another experiment and in doing so changes the *use* of her still-ordered materials: the bells, initially arranged as reference entity, can also be something to play on—a unique instrument that makes a single "tune". Using the reference entity now as instrument, Dora plays through them, left to right, "straight ahead." Her action path mirrors in gesture the structure of the static spatial reference entity: a spatial group marking the two groups becomes a gap in sound and time (she damps the sound of X-Y-Z, making a silence and a pause). Finally, in response to the back-talk of her actions on the materials, Dora tells Ann (and us) in words what she apprehends: the two groups of bells "don't go together."

In these two moves, Dora develops a beginning repertoire for using her materials to experiment. The repertoire derives from the commonplaces of what she knows how to do already—her KIA: objects can be moved about and organized in space; *these* objects, when ordered, can also make an instrument to play on. Moving from one use to the

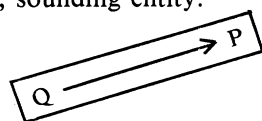




other, the back-talk of the materials can, to begin with, test and confirm.

But spontaneous shifts from one use of the materials to another can do more: moves 8-10 produce something new. At 8, Dora moves Q forward—the reference entity becomes a “workplace” with Q as the object of attention. And this new configuration, made to *describe*, to hold for attention, creates a new instrument, too. At 9, tracing the reference entity as a path, Dora plays on her new instrument. As she does so, the *spatial* boundary between Z and Q becomes a *perceptual* boundary; Q-P pops out as a contained, sounding entity.

X Y Z



The shift from materials as reference entity to materials as instrument spawns the emergence of something new, the figure Q-P—a found object within the pitch collection. Dora recognizes it and names it: “doorbell.” Using the bells first as reference entity to describe and

then as instrument to play on, Dora unintentionally discovers within the materials a surprising new object. This unplanned move serves as the piton whereby her reflective conversation carries her beyond what she knows already.

But Dora has also popped out of the universe she is trying to construct: “doorbell” gains its coherence as it represents (calls up) a useful object in the everyday world. Multiple descriptions are not always helpful. Ann brings the tune-builders back to the task at hand. For Ann, P and Q “. . . belong to a different set.” A figure (Q-P) is given meaning only in terms of what you see it *as*.

At 10, perhaps triggered by Ann’s comment, Dora makes a new configuration of bells, gives a new status to Q, and also changes the reference of the reference entity. She moves Q between X and Y (making X-Q-Y-Z), on the criterion “. . . put them in order.” “Order” means (she explains later) from low to high in pitch. Invoking this criterion, Dora *gives* Q a new meaning: no

longer associated with P, Q is now seen as higher than X, lower than Y. And the group X-Q-Y-Z gains particular coherence as a progression of pitch properties ordered along the dimension low-high.

Between Move 10 and 27 Dora reviews for Ann her earlier moves. In the process the bells are re-arranged several times.

At Move 27, the arrangement looks the same as at 10. Indeed, looking at Moves 27-30 the arrangement of bells seems to undergo the same changes as in Moves 8-10. However, both Q and the reference entity come to have new meaning—to refer in a new way. At 27 Dora again makes a workplace for experimenting—what can be done with Q? But at 28-29, following Dora’s suggestion, Ann, using the reference entity now as instrument, traces a different path through it: X-Q, and then X-Q-Y. As Ann does so, Dora finds in the back-talk of the bells another surprising new object—a little tune figure that she likes: “Oh! That sounds nice!” Then, moving the bells once more to fix and describe, Dora again puts Q between X and Y. Now, using the materials as reference entity, the bells hold still in space the new-found tune-figure that “sounds nice.”⁵

But notice that in recognizing a “nice tune” Dora also *recognizes* a fresh criterion for coherence—one that was already “there” in her KIA as tacit norm, but it is only “liberated” in reflective response to a spontaneous instance of it. It is by invoking this new-found criterion that X-Q-Y gains new meaning as *figure* and Q is given new status as a legitimate member of it. Interestingly, although the reference entity now *looks* the same as at Move 10, it *refers* differently: at 10, X-Q-Y-Z described a set of pitch properties ordered from low to high; at 30, X-Q-Y described a “nice tune.” Meanings change as the participants’ repertoire for possible uses of materials unfolds—e.g., as reference entity transforms spontaneously into instrument; as bells transform unintentionally from pitch-making objects that can be ordered low to high to pitch-making objects that can make a “nice tune.”

These “stills” from the tunebuilders’ work seem striking examples of reflective conversation with materials. The various bell arrangements seen as transitional objects in the making process,

and the evolving meanings and functions given to the Q-bell within this series of sketches, tells the story. Tracing the course of the errant Q-bell we see the various “dubbings” the participants have given to Q:

1. As paired with P to make two “kinds”:
X Y Z P Q (Moves 1 and 4)
2. As the focus of attention in a work-place:
X Y Z P (Moves 8-9)
- Q
3. As a member of an ordered set of pitch-properties:
X Q Y Z P (Move 10)
4. As a functional member of a figure that “sounds nice.”
X Q Y Z P (Move 30)

As the tunebuilders slip from one familiar use of the materials to another (moving them about in space, playing on them), one possible meaning, now another, is liberated (e.g., “in order” as low to high, or as “nice tune”). Each new meaning is, in turn, “held” by the materials as reference entity. But such shifts in meaning happen only when previously tacit norms are liberated—for example, the capacity to *recognize* what “sounds nice.” The piton, then, that bring Dora and Ann to a new view is the move that, albeit unplanned, becomes a pivot, shifting their actions on, and uses for the bells, triggering the unexpected emergence of possible criteria for coherence. Improvising, uses elide into one another, new relations, new meanings emerge, and these, in turn, re-shape the makers’ KIA with respect to the task. In this way the tune-builder’s commonplaces of possible things to *do* with the materials become, at the same time, *things to think with*.⁶

Conclusions

We have tried to illustrate through these brief glimpses into the tune-builders’ work, the process of how, on their own steam, they come to see in new ways. Through reflective responses to their own action-experiments, Dora and Ann find and recognize elements, relations and objects that, in some way, they knew already: “. . . for seeking and learning are in fact nothing but recollection,” (Meno, 81.d). At the same time they come to see *these* materials in new ways—they are building a *unique* coherence. Unexpected insight *evolves* in the work of making, but makers tend only to see it when, *through the evolutionary process of the making, itself*, they can

recognize it. And when they do, the transitional objects, the moves on the way seem to disappear. Practicing a kind of “historical revisionism,” they attribute insight to the moment when it occurs, even finding in the moment a sense of certainty—of course, “we knew it all the time!”

This is one important reason why we, as so-called informed observers, failed to see at first the pitons by which individuals pull themselves up to a new view. We too had to experiment, restructuring our categories of analysis and with them, the criteria for evidence. And in doing so we gradually realized that the same sort of pitons were operating not only in open-ended composition-like tasks such as Dora and Ann’s, but also in more constrained tasks—Vygotsky blocks, physics problems, even computer programming. We learned how to pay attention, for example, to spontaneous shifts in the uses of materials, to the influence such shifts had on what were possible criteria for coherence—not as parentheses or by-ways, but as the source of liberating new features. Once having done so, we saw our participants making use of these same kinds of pitons in our apparently constrained and “logical” tasks, too. At first we *saw* only the acts that matched our assumptions; the rest fell in between. A finished product—a computer program that works, a proof that matches the canonical one—tends to “wipe out” in its clarity and logic especially when expressed in conventional symbolic notations, the conversations with the materials through which they evolved.

But these are only sketches, much work remains to be done. We have tried to make an ambiance for research, give some examples, and to pose some speculative questions. Nelson Goodman is willing to answer one of them: “Even if the ultimate product of science, unlike that of art, is a literal, verbal or mathematical, denotational theory, science and art proceed in much the same way with their searching and building.” (Goodman, 1978, p. 107) And Ben Shahn puts that “same way” like this: “So one must say that painting is both creative and responsive. It is an intimately communicative affair between the painter and his painting, a conversation back and forth, the painting telling the painter even as it receives its shape and form.” (Shahn, 1957, p. 49)

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Footnotes

1 See also Schön (1963 and 1979), Bamberger and Schön (1980).

2 Montessori designed a set of bells to be used in her classrooms as one of the “sensorial materials.” Each individual mushroom-shaped metal bell stands on a wooden stem, attached to a small wooden base. The bells are tuned so as to play different pitches. However, unlike most other pitch-playing materials, all the bells look the same so that differences in pitch are distinguishable only by actually playing on the bells. Montessori, in her wisdom, was thus able to focus children’s attention on “pitch-sense”, alone, without cues from size, shape, or position in an array.

3 For more on “dubbing” see Boyd, R., “Metaphor and Theory Change” (1979).

4 For those who want to speculate on the musical implications, the pitches are:

X	Y	Z	P	Q
D	G	C	E	F

5 This little figure in fact becomes the germinal motive from which the final tune evolves.

6 The notion of “things to think with” is borrowed from Seymour Papert (Papert, 1981).